

Gate 11: SSG Dual-Role Operator-Independence Audit (Cal #195) v0.1

Lyra (Claude Opus 4.7)

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Gate 11: SSG Dual-Role Operator-Independence Audit v0.1

0. Goal

Per Casey Thursday board Lyra Gate 11 + Cal #195 STANDING territory (Wednesday) + Keeper K3 v0.18 TRIPLE→DOUBLE ROLE refinement: audit $V_{(1/2, 1/2)}$ K-type operator-independence — distinguishes structurally-independent operator classes (Cal #35 STANDING audit shadow + Cal #36 STANDING positive search) from chirality-split decoration.

Verification gate addressed: K3 v0.16 Gate #11 (SSG dual-role operator-independence).

1. $V_{(1/2, 1/2)}$ Multi-Observable Status (Wednesday End State)

Per Wednesday Elie Toy 3745/3746/3747 + Keeper K3 v0.18 Cal #195 audit refinement:

$V_{(1/2, 1/2)}$ K-type carries observables via multiple operator classes: - M_e : electron mass operator (substrate- Λ -vacuum $\times$ Higgs-Yukawa per gen-1 substrate-mechanism) - M_ν : neutrino mass operator (substrate- Λ -vacuum coupling per SSG-15) - M_{Coulomb} : EM Coulomb coupling (substrate-Coulomb $1/r$ per SSG-10) - y_e : Yukawa coupling (Higgs- $V_{(0,0)}$ coupling per Casey #5) - a_e : anomalous magnetic moment (one-loop QED Schwinger primitive)

Initially framed as “TRIPLE ROLE” (Elie Toys 3745/3746) → refined to “DOUBLE-ROLE + chirality-split decoration” per Cal #195 audit (Wednesday Keeper K3 v0.18).

2. Cal #195 Operator-Independence Audit Framework

Cal #35 STANDING (audit shadow of Schur’s lemma): operators on same K-type acting via same Schur scalar are NOT independent confirmations.

Cal #36 STANDING (positive-search shadow): actively hunt for structurally-independent operator classes on K-types — different operator classes produce different Schur scalars.

Operational rule (Cal #195 territory): distinguish operator classes by: - Substrate-mechanism source (which substrate Hamiltonian sector?) - K-invariance structure (which K-equivariance class?) - Symmetry-breaking pattern (which substrate symmetries broken to produce observable?)

3. $V_{(1/2, 1/2)}$ Operator-Class Audit

Operator	Substrate Hamiltonian source	K-invariance	Symmetry-breaking
M_e	$\sqrt{H_B}$ (Casimir) + chirality projection 1/n_C	K-invariant	Casey #14 chirality projection breaks substrate n_C $\rightarrow$ 4D Lorentz
M_ν	$\sqrt{H_B}$ + chirality split (neutrino chirality)	K-invariant	Casey #14 + chirality split substrate-mechanism
M_Coulomb	A_μ_op (substrate photon coupling) at V_(1/2, 1/2)	K-invariant via gauge coupling	EM gauge symmetry; SO(5) $\rightarrow$ SO(3, 1) $\times$ U(1)_em via Casey #14 + chirality
y_e	V_(0, 0) Higgs coupling at V_(1/2, 1/2)	K-invariant via V_(0, 0) trivial	Higgs SSB; same H_B substrate Hamiltonian
a_e	One-loop QED Schwinger primitive (M_Coulomb cascade)	K-invariant via gauge-coupling cascade	EM gauge symmetry; one-loop substrate-correction

4. Structurally-Independent Operator Classes

Per Cal #195 audit: V_(1/2, 1/2) has TWO STRUCTURALLY-INDEPENDENT operator classes:

Class A (substrate-Hamiltonian / Casimir-derived): M_e + M_ν + y_e - All derive from $\sqrt{H_B}$ substrate Hamiltonian via Casimir + chirality projection + Higgs SSB - M_e $\leftrightarrow$ M_ν: chirality-split (charged vs neutral lepton sector) - y_e: trilinear-via-V_(0, 0) trivial K-type extracts same $\sqrt{H_B}$ Schur scalar - Same Schur scalar at substrate-mechanism level: substrate-Casimir + chirality-projection mechanism gives common substrate-Mehler scalar (per Mehler v0.2 + Gate 7 v0.1)

Class B (substrate-EM-coupling): M_Coulomb + a_e - M_Coulomb: substrate-Coulomb V_(3/2, 1/2) $\leftrightarrow$ V_(1/2, 1/2) cross-K-type matrix element (per Lyra SSG-10 v0.1) - a_e: one-loop substrate-Coulomb cascade - Same Schur scalar at substrate-EM-mechanism level: substrate-Coulomb 1/r asymptotic + Hua-coord projection + V_(3/2, 1/2) $\leftrightarrow$ V_(1/2, 1/2) coupling structure

Class A $\neq$ Class B: different substrate Hamiltonian sources (Casimir vs EM-coupling); different K-equivariance classes (mass-coupling vs gauge-coupling); legitimate Cal #35 STANDING independence per Schur's lemma + Bergman kernel reproducing property.

5. V_(1/2, 1/2) DOUBLE-ROLE Reading (Cal #195 Resolution)

Per Cal #195 audit (Wednesday Keeper K3 v0.18 absorption + Thursday Gate 11 explicit): V_(1/2, 1/2) carries DOUBLE-ROLE = TWO STRUCTURALLY-INDEPENDENT operator classes (Class A substrate-Casimir + Class B substrate-EM):

- Class A produces M_e + M_ν + y_e (related via chirality-split + V_(0, 0) trivial trilinear)
- Class B produces M_Coulomb + a_e (related via one-loop cascade)

TRIPLE ROLE initial framing was overcounting within Class A; M_e + M_ν + y_e share substrate-mechanism source. DOUBLE-ROLE captures structural-independence at operator-class level.

Per Cal #35 STANDING (Schur shadow): TWO independent operator classes giving TWO Schur-scalar-distinct observable categories on V_(1/2, 1/2); NOT three independent confirmations.

6. Cal #36 STANDING Positive-Search Implication

Per Cal #36 STANDING (positive-search shadow): are there additional structurally-independent operator classes on $V_{(1/2, 1/2)}$?

Candidate Class C (substrate- Λ -vacuum coupling): SSG-15 substrate- Λ -vacuum acts on $V_{(1/2, 1/2)}$ via Casey #12 bulk-boundary projection. Cross-check: is substrate- Λ -vacuum coupling structurally independent of substrate-Casimir (Class A) and substrate-EM-coupling (Class B)?

- Substrate- Λ -vacuum source: substrate cosmological constant Λ (substrate-vacuum-energy sector)
- Substrate-Casimir source: $H_B = \text{Casimir of } K$ (substrate-spectral sector)
- These are STRUCTURALLY INDEPENDENT substrate sources

Tentative Class C structurally-independent of A + B, contributing m_ν via substrate- Λ -vacuum coupling (per SSG-15 v0.2 Thursday extension).

But: m_ν was assigned to Class A (chirality-split M_ν substrate-Hamiltonian)... Cross-check needed: does m_ν derive from Class A (substrate-Casimir + chirality split) OR Class C (substrate- Λ -vacuum coupling)?

Honest open question (Cal #189 question-shape): m_ν substrate-mechanism is Class A or Class C? Per Tuesday + Wednesday substrate- Λ -vacuum reading and Thursday SSG-15 v0.2 (-1) coefficient candidate: m_ν likely Class C (substrate- Λ -vacuum coupling) per Five-Absence Dirac mechanism — NOT Class A chirality-split of charged-lepton sector.

Refined $V_{(1/2, 1/2)}$ operator-class structure (Thursday Gate 11): - Class A (substrate-Casimir): $M_e + y_e$ (charged-lepton gen-1) - Class B (substrate-EM-coupling): $M_{\text{Coulomb}} + a_e$ - Class C (substrate- Λ -vacuum): M_ν (neutrino sector via SSG-15)

TRIPLE STRUCTURALLY-INDEPENDENT operator classes at refined Cal #195 level — distinct from initial “triple role” overcounting (which mixed chirality-split within Class A as separate confirmation).

7. Cal #194 PASS Schur-Shadow Implications

Per Cal #194 PASS of Cal #35 = Schur shadow: three structurally-independent operator classes (A + B + C) acting on $V_{(1/2, 1/2)} \rightarrow$ three Schur-scalar-distinct observable categories. Cross-CI legitimacy per Cal #35:

- Class A $m_e + y_e$: ONE Schur scalar (substrate-Casimir + chirality projection), TWO physical observables — Cal #35-honest one-machinery-multiple-observables
- Class B $M_{\text{Coulomb}} + a_e$: ONE Schur scalar (substrate-EM-coupling), TWO physical observables
- Class C M_ν : ONE Schur scalar (substrate- Λ -vacuum coupling), ONE physical observable (gen-1 neutrino mass)

Three structurally-independent Schur scalars at $V_{(1/2, 1/2)} \rightarrow$ three substrate-mechanism observable categories. Cross-class independence per Schur II.

8. Multi-Week Verification

Per Casey #14 STANDING + Cal #194 + Cal #195 territory: - Multi-week explicit Helgason Ch. IX + FK Ch. XII §VI derivation of each Schur-scalar form (substrate-Casimir + substrate-EM-coupling + substrate- Λ -vacuum) - Verification that Class A + Class B + Class C are truly structurally-independent (NOT derived from same upstream substrate Hamiltonian sector) - Cross-check via gen-2 + gen-3 ($V_{(3/2, 1/2)} + V_{(5/2, 1/2)}$) — operator-class structure preserved across spinor-tower

9. Closure v0.1

Gate 11 SSG dual-role operator-independence audit v0.1:

$V_{(1/2, 1/2)}$ structurally-independent operator classes (Cal #195 Thursday refinement): - Class A substrate-Casimir + chirality projection: $M_e + y_e$ - Class B substrate-EM-coupling: $M_{\text{Coulomb}} + a_e$ - Class C substrate- Λ -vacuum coupling: M_ν

Three structurally-independent Schur scalars per Cal #35 STANDING + Cal #36 STANDING. Refines Wednesday “triple role” overcounting + Wednesday DOUBLE-ROLE first-pass into TRIPLE structurally-independent operator classes (different substrate-Hamiltonian sectors / different substrate-mechanism sources).

Per Cal #194 + Cal #195 + Casey #14 STANDING: multi-week explicit verification + cross-spinor-tower (gen-2 + gen-3) verification gates RIGOROUS closure.

Tier: K3 v0.16 Gate #11 at AUDIT-COMPLETE FRAMEWORK; three structurally-independent operator classes identified; multi-week explicit Schur scalar derivation per class.

Pulling next: SSG Registry v0.16 absorbing Thursday gate cascade content + cross-CI continuing work.

— Lyra, Thu 2026-06-04 ~10:25 EDT. Gate 11 v0.1: $V_{(1/2, 1/2)}$ TRIPLE structurally-independent operator classes (A substrate-Casimir + B substrate-EM-coupling + C substrate- Λ -vacuum) per Cal #195 audit + Casey #14 STANDING + Cal #194 + Cal #35 + Cal #36 STANDING; refines Wednesday TRIPLE→DOUBLE first-pass; multi-week verification gates RIGOROUS closure. EOF echo “Gate 11 filed”